Confusion Matrix
Multivariate Metrics } 30 mins

Agenda

Hands on

⇒ Imbalanced class + Multiclass problem
⇒ Build classification Models.
⇒ SMOTE
⇒ Comparison of metrics Imbalance & Balance

⇒ Questions & Take away: for End of this Session

1. How are the assets frozen for deployment?
2. Multiple frameworks supporting deployment of ML models.
3. What is Streamlit?

Session 1 Revision

→ Regression:
Input - X (Numerical + categorical)
Out - 'Y' (Numerical continuous)

→ Classification:
Input - X (Numerical + Categorical)
Out - 'Y' (categorical Discrete)

⇒ What we saw?

Binary Classification. — Yes / No

Yesterday Extended

↳ Multiple classes — Yes / Maybe / No

ML/DL Goal

Data ? Preprocessing → What is consuming this data → What is the config → Performance?

Metrics

CHOICE of Model

Eg) Tree based?
Linear model?
Distance based?

Hyper param tuning

⇒ ① Logistic Regression - Binary class Model
(Because of S Sigmoid)

SOFTMAX Regression
(variant) ↳ Multi-Class classificat
(per class regression line)

Binary Metric | Multiclass Metr.

Matrix 2x2 | NxN

Outcomes TP/FP/TN/FN

loan approval.

Actuals

Low Med High

'' Actual

	App	Rej
Pred App	True App	false approval
Pred Rej	false Rej	True Rej

$$\text{Accuracy} = \frac{\text{True App} + \text{True Rej}}{\text{Total}}$$

Precision :- when I predict approved, how often is it actually approved?

$$\frac{\text{True Approved}}{\text{True App} + \text{false approval}}$$

Recall :- Out of actual approvals how many did the model actually find

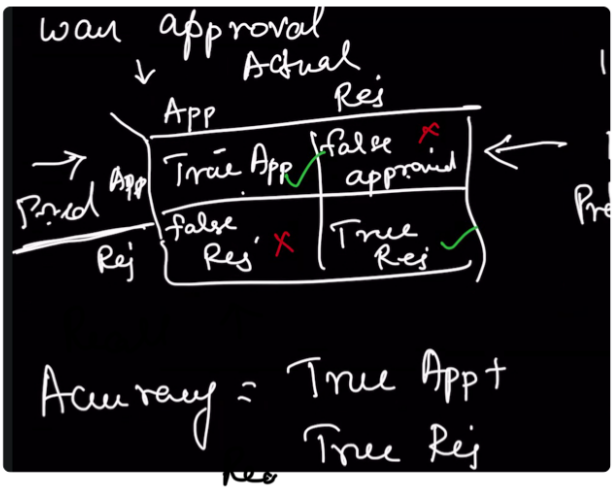
$$\frac{\text{True Approvals}}{\text{True approvals} + \text{false}\ \ldots}$$

Pred			
Low	True Low	False Lows	false Low
Med	False Med	True Med	False Med
High	false High	false High	True High

false High low

$C_{\text{low, High}}$

$$\text{Accuracy} = \frac{\text{Sum of diag}}{\text{Total}}$$



Every

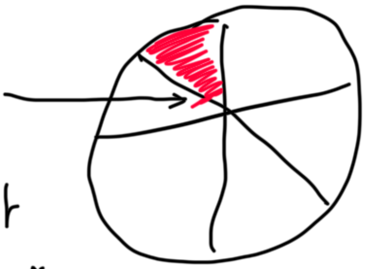
Predict

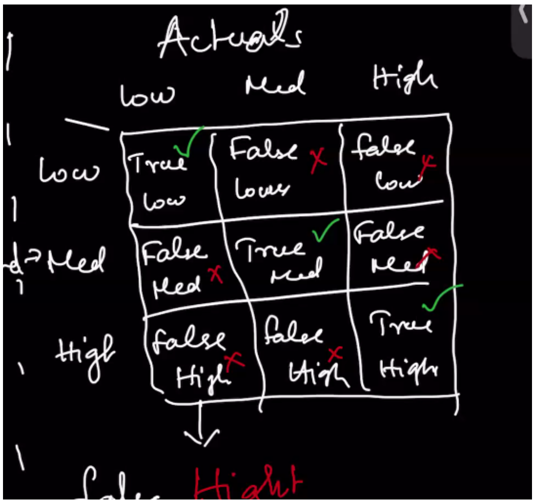
Precision

Rej.

Recall is a proportion against Your actuals

Precise weak metric against your prediction.





	Actuals Low	Med	High
Low	True Low	False Lows	False Low
Med	False Med	True Med	False Med
High	False High	False High	True High

False High

$$\text{Precision}_{low} = \frac{\text{True low}}{\text{True low} + \text{False lows } (C_{Med,low}, C_{Hig,low})}$$

$$\text{Precision}_{Med} = \frac{\text{True Med}}{\text{True}_{Med} + C_{LM} + C_{HM}}$$

$$\text{Precision}_{High} = \frac{\text{True High}}{\text{True High} + \text{false Highs.}}$$

$$\text{Recall}_{low} = \frac{\text{True lows}}{S(\text{Actual lows})}$$

$$\text{Recall}_{High} = \frac{\text{True Highs}}{S(\text{Actual High})}$$

$$\text{Recall}_{Med} = \frac{\text{True Med}}{S(\text{Actual Medium})}$$

Model → Predict Majority Well

IMBALANCED CLASSES — Data / Adjusting Metrics

In my target Y — A (80%) ⇒ Majority class; B (15%) ⇒ / C (5%) ⇒ Minority Class

Majority Head

Tail

long distribution

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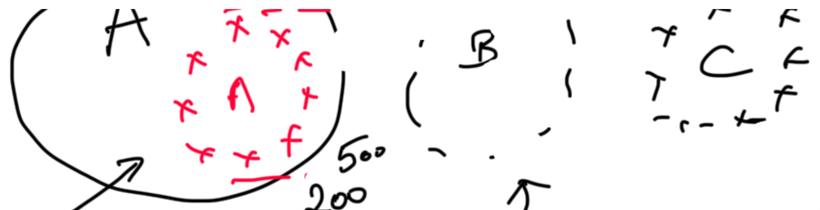
① UNDER SAMPLING

5000 200

70% Major
30% All Minors
100

UNDER SAMPLE But I make my



Reduce valueble information?

model learn all Classes Equally

X Random Undersamp
~~Nearmiss~~

Oversampling ⇒ Duplicates Minority

SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE

(1) Random Oversampling → Overfitting problem.

(2) SMOTE → New points KNN

Read on this self

a a

(3) ADASYN – Patterns Via / also Create duplicates of outliers from Miaory.

IMBALANCE Y

DT

		Actual 0	1 — fraudulent	
Prediction	Non-fraud 0	450	50	500
	fraud 1	45	5	50
		495	55	550

550
500 – 0
50 – 1

50 rows = 1
5 fraud

(1) $$\text{Accuracy} = \frac{450+5}{550} = \frac{455}{550} = 83.\%$$

null

The w…
in which I am quantifying my model perform has to change.

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② Can I somehow Engineer the data in such a way that I make my model understand fraudulent cases as well?

⇒ UNDERSAMPLING

× May reduce your models' understan on non-fraudulent cases

fraudulent 55 rows
Non-fraudulent 455 rows

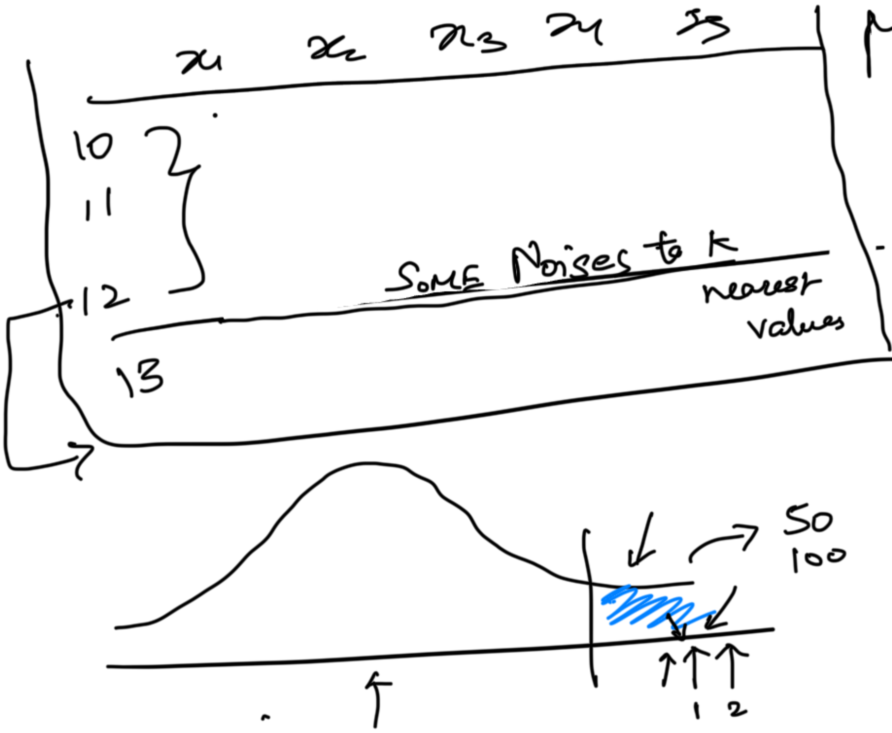
55 $\boxed{55, 55}$ ⇒ (110)

OVERSAMPLING

Oversample patterns in such a way that I increase the count of minor class

INCREASE COUNT OF MINOR CLASS

0 - 455
1 - 455



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70% Major — 60% A, 40% B

15% - B

15% - C

50 50

55 45

60 40

no rows — 70 30 → rows → 20% → 70%

30 30

How will we handle imbalance via model performance quantification?

1. Accuracy fails → favours class skewness

2. Precision & Recall methods.

Precision & Recall.

f_1 Score.

1. Arithmetic Mean
2. Exponent Moving Avg

(3) WMA

(4) Harmonic Mean.

$$\frac{\text{dividing total No. of values}}{\text{Sum of their reciprocal.}}$$

Driving 60 km $\xrightarrow{60\,km/h}$ $\xleftarrow[20\,km/h]{}$

$60 / 60\,km/hr = 1\,hr$

$60 / 20/hr = \underline{3\,hr}.$

Arith $= \frac{60+20}{2} = 40\,km/h$

$\underline{120\,kms} / \underline{4\,hours} = 30\,km/h$

$$H = \frac{2}{1/60 + 1/20} = \frac{2}{4/60} = 30\,km/hrs$$

Reciprocals → $1/60 + 1/20$

$$\boxed{H_{mean} = \frac{2ab}{a+b}}$$

It represents "True Rate" of a System

True Rate or True approvals out of all actual approvals — Recall

vs

True approvals predi...

f_1 Score $= \dfrac{2\ \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Avoid Risk

f_1 Medium Risk

out of all

Precision $\frac{\text{actual}}{\text{pred.}}$

F_1 High Risk

Muticlass Classification